

Problem 1:

Given a Binary Search Tree and a range [low, high]. Find all the numbers in the BST that lie in the given range. Note: Element greater than or equal to root go to the right side.

Concept: Binary Search Tree

Example 1:

Input:

```
      17
     /  \
    4    18
   /  \
  2    9
l = 4, h = 24
Output:
4 9 17 18
```

Example 2:

Input:

```
      16
     /  \
    7    20
   /  \
  1    10
l = 13, h = 23
Output:
16 20
```

Code:

```
#include <stdio.h>

#include <stdlib.h>

struct Node {
    int data;
    struct Node* left;
    struct Node* right;
};

struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->left = newNode->right = NULL;
    return newNode;
}
```

```

}

struct Node* insert(struct Node* root, int data) {
    if (root == NULL) {
        return createNode(data);
    }
    if (data < root->data) {
        root->left = insert(root->left, data);
    } else {
        root->right = insert(root->right, data);
    }
    return root;
}

void printInRange(struct Node* root, int low, int high) {
    if (root == NULL)
        return;
    if (root->data > low)
        printInRange(root->left, low, high);
    if (root->data >= low && root->data <= high)
        printf("%d ", root->data);
    if (root->data < high)
        printInRange(root->right, low, high);
}

int main() {
    struct Node* root = NULL;
    printf("Name : Kartika Patil\n");
    printf("PRN : 25070521219\n");
    printf("Section : D-2\n");
}

```

```

printf("Value Added Problem : 1\n");
printf("-----\n");
root = insert(root, 17);
insert(root, 4);
insert(root, 18);
insert(root, 2);
insert(root, 9);
int low = 4, high = 24;
printf("Output: ");
printInRange(root, low, high);
return 0;
}

```

Output:

main.c	Output
<pre> 39 if (root->data < high) 40 printInRange(root->right, low, high); 41 } 42 43 int main() { 44 struct Node* root = NULL; 45 46 printf("Name : Kartika Patil\n"); 47 printf("PRN : 25070521219\n"); 48 printf("Section : D-2\n"); 49 printf("Value Added Problem : 1\n"); 50 printf("-----\n"); 51 52 root = insert(root, 17); 53 insert(root, 4); 54 insert(root, 18); 55 insert(root, 2); 56 insert(root, 9); 57 58 int low = 4, high = 24; 59 60 printf("Output: "); 61 printInRange(root, low, high); 62 63 return 0; 64 } </pre>	<pre> Name : Kartika Patil PRN : 25070521219 Section : D-2 Value Added Problem : 1 ----- Output: 4 9 17 18 === Code Execution Successful === </pre>